

Disease and Symptoms Dataset

This Dataset consists of Categorical variables of Different Diseases with their symptoms

Attribute Function:

- Disease - Target Variable
- Symptom_1 to Symptom_17

```
In [1]: # Importing Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings("ignore")
from sklearn.preprocessing import MultiLabelBinarizer, LabelEncoder
```

```
In [2]: # Load the dataset
data = pd.read_csv("C:\\Users\\rache\\OneDrive\\Rachel Study Materials\\Project Dataset\\Disease_and_Symptoms_Diagnostic.csv")
```

```
In [3]: # First Five rows of the dataset
data.head()
```

```
Out[3]:
```

	Disease	Symptom_1	Symptom_2	Symptom_3	Symptom_4	Symptom_5	Symptom_6	Symptom_7
0	Fungal infection	itching	skin_rash	nodal_skin_eruptions	dischromic_patches	NaN	NaN	NaN
1	Fungal infection	skin_rash	nodal_skin_eruptions	dischromic_patches	NaN	NaN	NaN	NaN
2	Fungal infection	itching	nodal_skin_eruptions	dischromic_patches	NaN	NaN	NaN	NaN
3	Fungal infection	itching	skin_rash	dischromic_patches	NaN	NaN	NaN	NaN
4	Fungal infection	itching	skin_rash	nodal_skin_eruptions	NaN	NaN	NaN	NaN

```
In [4]: #Shape of the dataset
data.shape
```

```
Out[4]: (304, 18)
```

```
In [5]: # datatypes
data.dtypes
```

```
Out[5]: Disease      object
Symptom_1      object
Symptom_2      object
Symptom_3      object
Symptom_4      object
Symptom_5      object
Symptom_6      object
Symptom_7      object
Symptom_8      object
Symptom_9      object
Symptom_10     object
Symptom_11     object
Symptom_12     object
Symptom_13     object
Symptom_14     object
Symptom_15     object
Symptom_16     object
Symptom_17     object
dtype: object
```

```
In [6]: # Information of the Dataset
data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 304 entries, 0 to 303
Data columns (total 18 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Disease          304 non-null    object
1   Symptom_1        304 non-null    object
2   Symptom_2        304 non-null    object
3   Symptom_3        304 non-null    object
4   Symptom_4        272 non-null    object
5   Symptom_5        234 non-null    object
6   Symptom_6        186 non-null    object
7   Symptom_7        158 non-null    object
8   Symptom_8        140 non-null    object
9   Symptom_9        120 non-null    object
10  Symptom_10       110 non-null    object
11  Symptom_11       68 non-null     object
12  Symptom_12       47 non-null     object
13  Symptom_13       30 non-null     object
14  Symptom_14       19 non-null     object
15  Symptom_15       18 non-null     object
16  Symptom_16       10 non-null     object
17  Symptom_17       1 non-null      object
dtypes: object(18)
memory usage: 42.9+ KB

```

```

In [7]: # Summary of the dataset
data.describe()

```

Out[7]:

	Disease	Symptom_1	Symptom_2	Symptom_3	Symptom_4	Symptom_5	Symptom_6	Symptom_7
count	304	304	304	304	272	234	186	158
unique	41	34	48	54	50	38	32	26
top	Hepatitis D	vomiting	vomiting	fatigue	high_fever	headache	nausea	abdominal_pain
freq	10	46	56	46	30	24	21	22

- From the summary of the dataset Disease has 41 unique values.
- This dataset consist of Null Values.

```

In [8]: data.isnull().sum()

```

```

Out[8]: Disease          0
Symptom_1          0
Symptom_2          0
Symptom_3          0
Symptom_4          32
Symptom_5          70
Symptom_6         118
Symptom_7         146
Symptom_8         164
Symptom_9         184
Symptom_10        194
Symptom_11        236
Symptom_12        257
Symptom_13        274
Symptom_14        285
Symptom_15        286
Symptom_16        294
Symptom_17        303
dtype: int64

```

Data Preprocessing

```

In [9]: # Symptom columns
symptom_cols = [f'Symptom_{i}' for i in range(1, 18)]

# Convert symptom columns into list
data['symptom_list'] = data[symptom_cols].values.tolist()

# Remove NaN values
data['symptom_list'] = data['symptom_list'].apply(
    lambda x: [symptom for symptom in x if pd.notna(symptom)]
)

```

```

In [10]: # Encode symptoms

```

```
mlb = MultiLabelBinarizer()

symptom_encoded = pd.DataFrame(
    mlb.fit_transform(data['symptom_list']),
    columns=mlb.classes_
)
```

```
In [11]: # Encode target variable (Disease)
le = LabelEncoder()

symptom_encoded['Disease'] = le.fit_transform(data['Disease'])
```

```
In [12]: # Save final encoded dataset
symptom_encoded.to_csv("encoded_disease_dataset.csv", index=False)

symptom_encoded.head()
```

```
Out[12]:
```

	abdominal_pain	abnormal_menstruation	acidity	acute_liver_failure	altered_sensorium	anxiety	back_pain	bel
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0

5 rows × 132 columns



```
In [13]: df = pd.read_csv("encoded_disease_dataset.csv")
```

```
In [14]: df.shape
```

```
Out[14]: (304, 132)
```

- In this we used Label encoder to encode the target variable and Multilabel Binarizer for Symptoms columns.
- The Dataset is ready for Model Training.